

FP Phase 1 - Project Plan, describe datasets, joins, tasks, and metrics

- Due Nov 3, 2025 by 3:59pm
- Points 50
- Submitting a file upload
- Available Oct 20, 2025 at 3pm - Nov 4, 2025 at 12:59pm

This assignment was locked Nov 4, 2025 at 12:59pm.

FP Phase 1 - Project Plan, describe datasets, joins, tasks, and metrics

Note: By design, every aspect of the final project is up to the interpretation of your team. Don't think of the objectives as a checklist!

The module Final Project - Flight Delays brings together all material related to the final project (starter notebook, datasets, etc.), detailing the schedule, project description, and expectations (look at each phase submission for the rubric that will be used for grading that submission). The project will be completed in several phases as described below.

Project phases must be completed and submitted on a group basis.

This webpage is a good reference on how to write a good abstract: <https://www.aje.com/arc/make-great-first-impression-6-tips-writing-strong-abstract/>  (<https://www.aje.com/arc/make-great-first-impression-6-tips-writing-strong-abstract/>)

Objectives

1. Explain data (i.e., simple exploratory analysis of various fields, such as the semantic as well the intrinsic meaning of ranges, null values, categorical/numerical, mean/std.dev to normalize and/or scale inputs). Identify any missing or corrupt (i.e., outlier) data.
2. Define the outcome (i.e., the evaluation metric and the target) precisely, including mathematical formulas.
3. How do you ingest the CSV files and represent them efficiently? (think about different file formats)
4. Summarized relevant datasets **based on three months of the joined On Time Performance and Weather (OTPW data)**
5. Checkpoint the data after any operation to avoid wasting time and resources!
6. Split the data train/validation/test - making sure no leaks occur, for example: normalize using the training statistics.
 1. HINT: see Additional resources 2 below: Cross-validation in Time Series (very different from regular cross-validation)

SUBMISSION SUGGESTIONS:

1. You will submit a single phase 1 project report (in the form of a Databricks notebook) per group (please nominate one team member to do this), which must contain the following information (**please see the Rubric for this phase to see how your submission will be graded**). **Very important: You shouldn't have ANY code in the reports that you submit. Code notebooks that you use should be apart from the report notebook, and should be shared as References at the end (you can share the link).**
 - Project title
 - Phase leader plan
 - Credit assignment plan (who does/did what)
 - Project abstract. Try to be concise.
 - Data you plan to use
 - Some summary visual EDA
 - Which machine learning algorithm(s) are you considering using and why?
 - Metrics that you might use to measure success (standard metrics and domain-specific metrics)
 - A block diagram (Gantt diagram) of the key steps involved in completing this task and a timeline
 - Description of the pipeline steps you plan to use
 - Provide team members' names, emails, and photos
 - Indicate which part of the project each team member will be working on. e.g.,
 - Jane plans to build text-based features based on the product description and optimize random forest models using all available features and work on precision-recall measures;
 - Naveen might sign up for processing all the numerical data (standardizing, dealing with missing values) and optimize regularized logistic regression models using all available features, and work on F-measure metrics.
 - As a team, you will need to integrate your pipelines, your results, your discussion, and your conclusions into one notebook. This is relatively straightforward in a collaborative notebook such as DataBricks.
 - **Please see the Rubric for this phase to see how your submission will be graded.**
 - Conclusion and next steps
 - Open issues or problems
2. Phase leader plan in Table format (**please insert the table into the top of your Jupyter notebooks**)
 1. Each person assumes the project manager for one week (4 work weeks for the project)
 2. E.g., Phase 1: Project Plan, describe datasets, joins, tasks, and metrics | John Doe, etc... (please put in table format)
3. Credit assignment plan (who does/did what) in Table format (**please insert the table into the top of your Jupyter notebooks**)

1. Set up the tasks to be completed by phase; estimate the effort in terms of person-hours to complete the task, and assign the person responsible for that task
2. A good idea will be to set up SMART goals. S.M.A.R.T. is a mnemonic acronym, giving criteria to guide in the setting of goals and objectives for better results, for example in project management, employee-performance management, and personal development. For more background, please see: <https://www.atlassian.com/blog/productivity/how-to-write-smart-goals>. 
(<https://www.atlassian.com/blog/productivity/how-to-write-smart-goals>)
3. For ideas on how to create and manage such a table see the following:
 1. <https://support.google.com/a/users/answer/9310193?hl=en> 
(<https://support.google.com/a/users/answer/9310193?hl=en>)
4. Remember to keep this credit assignment plan up-to-date so you get the credit!

Please submit all documents in HTML format, with the following title format:

You will submit the following item:

1. DataBricks notebook in .html format
 - It is important to upload this file separately.
 - Please make sure the links/images are visible to others by viewing your file in incognito mode on your browser
 - Your submission will be penalized if you do NOT meet these requirements.

Project Phase 1 (1)

Criteria	Ratings		Pts
Abstract A good abstract should start defining the objective of the project. Then, it should define the data used as well as the source of the data. You can mention any interesting fact from your EDA, as long as it's clear and concise. Then, you will define the metric that you will used for the project. Because this is a planning phase, the second paragraph should be in future tense, explaining the Machine Learning algorithms that you plan to use, as well as any relevant information of your plan. Be concise!	10 to >0.0 pts Full Marks	0 pts No Marks	10 pts
Data description Basic analysis and understanding of data. For any calculation, use the 3 month sample dataset. 1.- Data dictionary (some datasets have hundreds of records, so focus on the data that passes your missing value analysis). 2.- Data size and source. 3.- How are you digesting the data and your checkpointing strategy. 4.- Missing and duplicate value plan and analysis. 5.- Quick and dirty EDA (summary statistics, etc.)	5 to >0.0 pts Full Marks	0 pts No Marks	5 pts
Machine algorithms and metrics. Description of algorithms to be used (list the names, implementations, loss functions that you will use) Description of metrics and analysis to be used (be sure to include the equations for the metrics) Here you will explain if you decided to do a Regression or a Classification problem and why. Plan the algorithms that you will plan to use for the project.	10 to >0.0 pts Full Marks	0 pts No Marks	10 pts
Machine Learning Pipelines Create your ML pipeline: from data ingestion to model. You NEED to create a block diagram and attach it in the notebook.	5 to >0.0 pts Full Marks	0 pts No Marks	5 pts
Overall proposal appearance Jupyter notebook Block diagrams	5 to >0.0 pts Full Marks	0 pts No Marks	5 pts

Criteria	Ratings		Pts
Gantt chart Layout (headings) Important thing: You always submit a report! So it should NOT have any code!			
Phase leader plan in Table format Phase leader plan in Table format No leader plan no grade No leader plan in table format no grade	5 to >0.0 pts Full Marks	0 pts No Marks	5 pts
Credit assignment plan Credit assignment plan (who does/did what) in Table format No Credit assignment plan means ZERO points Credit assignment plan not in Table format means ZERO points	10 pts Full Marks	0 pts No Marks	10 pts
Total Points: 50			